

BOROUGH OF MANHATTAN COMMUNITY COLLEGE

City University of New York

Department of Computer Information Systems**Office F930/Phone: 212-220-1476****Title of Course: Database Systems I****CIS 395-1100****Spring 2026****Class hours: 2****Lab hours: 3****Credits : 3****Instructor:** Younes Benkarroum**Email:** ybenkarroum@bmcc.cuny.edu**Office Hours:** Monday 2:00pm – 4:00pm**Office Number:** F-1030L (Phone Extension 7227)**Course Description:**

This course discusses the background of designing, implementing and operating database management systems. It covers conceptual data modeling and the entity relational data model in the beginning. Then, the relational data model is the focus in this course. Students learn abstract languages such as relational algebra and their commercial implementations such as Structured Query Language (SQL). Database design technique, concepts of data integrity, relational normalization theory, security, privacy, and concurrence control are introduced.

Basic skills: ENG 088; ESL 054; ACR 094; MAT 012, 014, 041, or 051**Prerequisites:** CSC 110 or CSC 111 or Departmental approval

Course Student Learning Outcomes (Students will be able to...)	Measurements (means of assessment for student learning outcomes listed in first column)
1. Define database concepts and terms.	1. Short essay questions in exams, SQL programming questions in exams and data application project.
2. Explain the physical data organization and the indexing of a database.	2. Short essay questions in exams.
3. Apply the principle of system analysis in the real world for database design.	3. Database application project with documentation.
4. Analyze the database system specification.	4. Database application project with documentation.
5. Construct the database for a major application project.	5. Database application projects with documentation.
6. Evaluate and refine the designs of conceptual data modeling.	6. Short essay questions in exams and database application project.

Below are the college's general education learning outcomes, the outcomes that are checked in the left-hand column indicate goals that will be covered and assessed in this course.

	General Education Learning Outcomes	Measurements (means of assessment for general education goals listed in first column)
☐	Communication Skills- Students will be able to write, read, listen and speak critically and effectively.	
☒	Quantitative Reasoning- Students will be able to use quantitative skills and the concepts and methods of mathematics to solve problems.	Use formulas and concepts of mathematics to solve problems in SQL programming assignments.
☐	Scientific Reasoning- Students will be able to apply the concepts and methods of the natural sciences.	
☐	Social and Behavioral Sciences- Students will be able to apply the concepts and methods of the social sciences.	

☐	Arts & Humanities- Students will be able to develop knowledge and understanding of the arts and literature through critiques of works of art, music, theatre or literature.	
☒	Information & Technology Literacy- Students will be able to collect, evaluate and interpret information and effectively use information technologies.	Use a high-level computer SQL programming language to create database application software.
☐	Values- Students will be able to make informed choices based on an understanding of personal values, human diversity, multicultural awareness and social responsibility.	

Required Text & Readings:

Textbook: Database Systems: A practical Approach to Design, Implementation and Management 6th ed
Author: Thomas Connolly and Carolyn Begg
Publisher: Addison Wesley, 2015
ISBN-13: 978-0132943260

Other Resources: Flash/USB drive is recommended.

Outline of Topics:

- Introduction to Databases
- The Relational Model
- The Relational Algebra
- The Relational Calculus
- SQL: Data Manipulation
- SQL: Data Definition
- Web Technology and Database Management Systems (Project)
- Entity-Relation Modeling
- Enhanced Entity-Relation Modeling
- Normalization

Evaluation & Requirements of Students:

Midterm exam	25%
Final exam	25%
Labs	20%
Homework	10%
Project	20%
Total:	100%

Class Participation

Participation in the academic activity of each course is a significant component of the learning process and plays a major role in determining overall student academic achievement. Academic activities may include, but are not limited to, attending class, submitting assignments, engaging in in-class or online activities, taking exams, and/or participating in group work. Each instructor has the right to establish their own class participation policy, and it is each student's responsibility to be familiar with and follow the participation policies for each course.

BMCC is committed to the health and well-being of all students. It is common for everyone to seek assistance at some point in their life, and there are free and confidential services on campus that can help.

Single Stop www.bmcc.cuny.edu/singlestop, room S230, 212-220-8195, singlestop@bmcc.cuny.edu. If you are having problems with food or housing insecurity, finances, health insurance or anything else that might get in the way of your studies at BMCC, come by the Single Stop Office for advice and assistance. Assistance is also available through the Office of Student Affairs, S350, 212-220-8130, studentaffairs@bmcc.cuny.edu.

Counseling Center www.bmcc.cuny.edu/counseling, room S343, 212-220-8140, counselingcenter@bmcc.cuny.edu. Counselors assist students in addressing psychological and adjustment issues (i.e., depression, anxiety, and relationships) and can help with stress, time management and more. Counselors are available for walk-in visits.

Office of Compliance and Diversity <https://www.bmcc.cuny.edu/about-bmcc/compliance-diversity>, room S701, 212-220-1236. BMCC is committed to promoting a diverse and inclusive learning environment free of unlawful discrimination/harassment, including sexual harassment, where all students are treated fairly. For information about BMCC's policies and resources, or to request additional assistance in this area, please visit or call the office, or email olevy@bmcc.cuny.edu, or twade@bmcc.cuny.edu. If you need immediate assistance, please contact BMCC Public safety at 212-220-8080.

Office of Accessibility www.bmcc.cuny.edu/accessibility, room N360 (accessible entrance: 77 Harrison Street), 212-220-8180, accessibility@bmcc.cuny.edu. This office collaborates with students who have documented disabilities, to coordinate support services, reasonable accommodations, and programs that enable equal access to education and college life. To request an accommodation due to a documented disability, please visit, call the office or email.

BMCC Policy on Plagiarism and Academic Integrity Statement

Plagiarism is the presentation of someone else's ideas, words or artistic, scientific, or technical work as one's own creation. Using the idea or work of another is permissible only when the original author is identified. Paraphrasing and summarizing, as well as direct quotations, require citations to the original source. Plagiarism may be intentional or unintentional. Lack of dishonest intent does not necessarily absolve a student of responsibility for plagiarism. Students who are unsure how and when to provide documentation are advised to consult with their instructors. The library has guides designed to help students to appropriately identify a cited work. The full policy can be found on BMCC's Web site, www.bmcc.cuny.edu. For further information on integrity and behavior, please consult the college bulletin (also available online).

Supplemental Instruction (SI)

BMCC is committed to student success; many foundational and gateway courses, including this course, utilize Supplemental Instruction (SI). SI is a free, voluntary academic success program that offers study sessions led by trained SI Leaders. Students are strongly encouraged to attend at least five out-of-class SI sessions during the semester, as institutional data indicate that students who reach this level of participation experience pass rates that are approximately 10–30% higher than those who do not attend SI sessions. Students who attend more than five sessions demonstrate even higher pass rates, making SI a valuable resource for academic success.

SI Leaders are paid student advocates who have mastered the course content and have been trained to facilitate collaborative group sessions. They attend class meetings and schedule two out-of-class small-group sessions per week to review course material, discuss key concepts, develop effective study strategies, and prepare for exams and assignments. Please note: SI Leaders do not grade student work or influence course grades. If you have questions about how SI works in this class, please contact the SI Leader or the professor. The SI Leader for this course is Jayden Tan.